

Binary outcomes, logit, probit, and likelihood

Slide route: Lecture11v2.tex: LPM, nonlinear models, effects, MLE and model selection. Chapter 11. Core route plus assigned extensions.

Session 15 | Model a probability

Outcomes: Interpret an LPM coefficient, calculate a logit probability and marginal effect, explain likelihood, and distinguish odds from probabilities.

100-minute core route

0–10	Why a binary outcome has a conditional probability as its mean.
10–25	LPM and its properties.
25–45	Logit/probit links and predicted probabilities.
45–55	Pause and coefficient interpretation.
55–75	Marginal effects, discrete changes, odds.
75–90	Bernoulli likelihood and estimation intuition.
90–100	One application table and exit ticket.

This is a dense deck. Assign repeated coefficient screenshots, the extended MLE algebra, and detailed model-selection slides for reading. Revisit AIC/BIC and model comparison in session 16. Do not try to narrate every slide.

Worked example | LPM versus logit

Use with slides Lecture11v2 slides 4–16: Binary Choice Models, Linear Probability Model, Properties of the LPM, Nonlinear Models, and Logit Model. Use the LPM range problem on slides 6–11 and the logit probability calculation on slides 12–16.

For a synthetic LPM $\hat{p} = .2 + .1X$, raising X by one gives a ten-percentage-point change. At $X = 9$, it predicts 1.1, illustrating the range problem. Binary errors have conditional variance $p(X)[1 - p(X)]$, so heteroskedasticity is natural.

$$p(X) = \Lambda(-2 + .5X) = \frac{1}{1 + \exp(2 - .5X)}.$$

At $X = 4$, $p = .5$. At $X = 5$, $p = \Lambda(.5) \simeq .6225$. The one-unit discrete change is .1225, or 12.25 percentage points.

$$\frac{\partial p}{\partial X} = .5p(1 - p); \quad \text{at } X = 4, \quad .5(.5)(.5) = .125.$$

The derivative and finite change are close, not identical.

Interpretation, probit, and likelihood

Use with slides Lecture11v2 slides 17–30 for probit/logit coefficient interpretation, marginal effects, and odds; slides 31–39 for MLE and the Bernoulli likelihood.

For the logit example, a one-unit X increase multiplies the odds by $e^{.5} \simeq 1.649$. This is not a 64.9-percentage-point rise in probability.

Probit If $p(X) = \Phi(\beta_0 + \beta_1 X)$, its derivative is $\phi(\beta_0 + \beta_1 X)\beta_1$. Both logit and probit coefficients move an index; compare predicted probabilities or marginal effects rather than raw coefficient magnitudes across links.

Binary regressor For a dummy D , report a discrete probability change between $D = 0$ and $D = 1$ at specified other covariates, or average those individual changes. A derivative is not the natural one-unit effect for a binary regressor.

Board example | Bernoulli likelihood

For independent observations $y = (1, 0, 1)$ and a common success probability p :

$$L(p) = p^2(1 - p), \quad \ell(p) = 2 \log p + \log(1 - p).$$

$$\ell'(p) = 2/p - 1/(1 - p) = 0 \quad \Rightarrow \quad \hat{p} = 2/3.$$

For regression, replace the common p with $p_i = F(X_i' \beta)$:

$$\ell(\beta) = \sum_i \{y_i \log p_i + (1 - y_i) \log(1 - p_i)\}.$$

Software maximizes this expression numerically. The objective rewards high probability for observed outcomes; it does not minimize squared residuals.

Practical caution Complete separation can prevent finite logistic MLE coefficients. Tiny standard errors or convergence messages should be read, not ignored.

Exit answers An LPM coefficient .03 means three percentage points per unit. A logit coefficient .03 is not generally three percentage points. Average marginal effects average observation-specific effects; effects at mean covariates are a different calculation.

Preparation For session 16, bring a one-paragraph causal question and propose an estimator with its key assumption. Read the remaining model-selection slides; do not treat classification accuracy as evidence of causality.

Source: Lecture11v2.tex. Numerical examples are synthetic; extended MLE/model selection are assigned preparation for review.